Ujjwal Pawar

Ph.D. Researcher in 5G/6G Mobile Networks, Open RAN and Telco Cloud

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SUMMARY

Ph.D. candidate at the University of Edinburgh whose research focuses on reducing the operating costs of mobile networks, with industry R&D experience at Nokia and Keysight Technologies. My work spans the full stack of real-world 5G systems: I co-built and operate a 20-cell campus-wide 5G Open RAN network, and use it to build systems for cloud-based RAN emulation, automated fault detection and diagnosis, network-side device localisation, and elastic, energy-efficient use of compute. I take ideas from design through prototype to evaluation on live networks, and my work has appeared at ACM MobiCom, ACM SIGCOMM CCR, ACM HotMobile and IEEE GLOBECOM, including a Best Demo Award at ACM MobiCom '25.

RESEARCH INTERESTS

Open RAN and RAN Intelligent Controllers • Telco Cloud and vRAN Compute Efficiency • AI-RAN • Network Automation and Observability • Cellular Positioning and Sensing • 6G

SKILLS

- **Wireless and RAN:** 5G NR, 6G, 3GPP RAN1 and RAN2 Specifications, Link Adaptation, Radio Resource Management, Mobility and Handover, Open RAN (O-RAN), CU/DU/RU Disaggregation, O-RAN 7.2x Fronthaul, FAPI, 5G Core
- **RIC and Automation:** Near-RT RIC, xApp Development (E2SM-KPM, E2SM-RC), O-RAN SC RIC, FlexRIC, Closed-Loop Control, Anomaly Detection, Fault Diagnosis, Policy-Constrained Automation
- **Telco Cloud and Infrastructure:** Kubernetes, Helm, Containers, NFV, vRAN and Cloud RAN, Real-Time Linux Tuning, CPU Isolation, DPDK, SR-IOV, Arista Switching (10/25/100 GbE), PTP Time Synchronisation, Public Cloud, NVIDIA GH200
- **Positioning and Measurements:** Multi-Cell 5G NR Localisation, Uplink SRS Processing, Time-of-Flight, Angle-of-Arrival, TDoA, Kalman Filtering, Radio Measurement Campaigns
- **Platforms and Programming:** Python, C/C++, Git, srsRAN, OpenAirInterface (OAI), OCUDU, USRP Software-Defined Radios, Telegraf, InfluxDB, Grafana, AI-Assisted (Agent-Based) Coding Workflows

PROFESSIONAL EXPERIENCE

AI/ML Research Intern – Open RAN and 6G Communications
Keysight Technologies, Edinburgh, UK | Sep 2026 – Present

Research Assistant
Netsys Lab, School of Informatics, University of Edinburgh, Edinburgh, UK | Jan 2025 – Jun 2025

- Built the compute and network infrastructure for a 20-cell 5G Open RAN network: 20 multi-vendor radio units across 6 campus sites, an Arista 100G fronthaul switch, PTP timing from a GPS-referenced grandmaster, and a Kubernetes platform optimised for real-time RAN workloads.
- Deployed E2-based near-RT RICs (O-RAN SC RIC, FlexRIC) with the KPIMON xApp, and automated multi-node deployment of RAN, core and RIC functions with Kubernetes and Helm.
- Ran a 15 km walking measurement campaign collecting about 11,000 GPS-tagged RSRP, RSRQ and SINR samples, used to build measurement-based coverage maps.

System Specification Engineer
Nokia R&D, Bangalore, India | Jan 2022 – Sep 2022

- Designed and specified Link Adaptation algorithms based on 3GPP RAN1 and RAN2 specifications to improve Voice over New Radio (VoNR) performance in 5G NR.

Research Assistant
Indian Institute of Technology Hyderabad, Hyderabad, India | Jan 2019 – Dec 2021

- Designed a traffic-aware compute resource tuning scheme that lets Cloud RAN servers run at lower CPU frequencies during off-peak hours, reducing energy use and fronthaul bandwidth (IEEE GLOBECOM 2021).

- Profiled the energy consumption of RAN functional splits on a real-time OpenAirInterface testbed with USRP software-defined radios (IEEE 5GWF 2020).
- Co-developed a traffic-prediction-based scheme that consolidates Cloud RAN processing onto up to 18% fewer servers without overload (IEEE NetSoft 2022).

RESEARCH PROJECTS

- **Campus5G – Campus-Scale Private 5G Open RAN Testbed (Co-Lead):** The first campus-wide O-RAN-compliant private 5G network, covering 50 acres of the University of Edinburgh with 20 radio units from two vendors, and a platform for research on RAN, RICs and AI-driven applications. (ACM SIGCOMM CCR 2025; Best Demo Award, ACM MobiCom '25)
- **Chronos – Cloud-Based RAN Emulation (Lead):** A RAN emulator that runs unmodified RAN software on the public cloud, emulating 10,000 base stations and 10,000 devices on 130 cloud instances by using virtual time to preserve real-time RAN behaviour. Includes a closed-loop xApp that tunes the link adaptation target BLER through E2, with measured BLER tracking the target (5.1% → 23.8% → 4.9%). (ACM MobiCom 2026; ACM HotMobile 2025)
- **Distributed 5G NR Localisation (Under Review):** A network-side system that locates devices from uplink reference signals received at multiple synchronised base stations, combining time-of-flight, angle-of-arrival and time-difference measurements, with 3.6 m median error over six months and 50 km of real-world trajectories.
- **Elastic vRAN Baseband (Under Review):** A system that resizes running vRAN baseband workloads and relocates them between servers without traffic loss or measurable service disruption, so compute can follow demand instead of being provisioned for peak.
- **Panoramic – Cross-Layer Observability for Open RAN (Lead):** A measurement platform that correlates RAN, 5G core, Kubernetes, server and switch telemetry for fault diagnosis, with an AI agent that runs remediation under tiered approval policies. (ACM Open AI-RAN Workshop 2026)
- **SpotLight – Explainable Anomaly Detection for Open RAN:** An edge–cloud anomaly detection system over 600+ RAN and platform metrics, with 13% higher F1 score and 4–7× lower bandwidth than baselines. (ACM MobiCom 2024)
- **AI-RAN on NVIDIA GH200 (Ongoing):** Exploring how AI workloads and latency-critical RAN functions can share the same CPU–GPU infrastructure without interfering with each other.

EDUCATION

Ph.D. in Computer Science and Engineering
University of Edinburgh, School of Informatics, Edinburgh, UK | Oct 2022 – Present
Supervisor: Prof. Mahesh K. Marina.

M.Tech in Computer Science and Engineering (CGPA: 9.19/10)
Indian Institute of Technology Hyderabad, Hyderabad, India | Jan 2019 – Dec 2021
Thesis: Traffic-Aware Compute Resource Tuning of vRANs for Energy Efficient 5G Networks.

Bachelor of Engineering in Computer Science and Engineering (CGPA: 7.3/10)
University Institute of Technology, RGPV, Bhopal, India | Aug 2014 – May 2018

PUBLICATIONS

Conference and Journal Papers

- **U. Pawar***, A. E. Ferguson*, Y. Xia, X. Foukas, M. K. Marina, and B. Radunovic, "Chronos: Scalable and Cost-Effective Cloud-based RAN Emulation System," ACM MobiCom, 2026.
- A. E. Ferguson*, **U. Pawar***, T. Wang, and M. K. Marina, "Campus5G: A Campus Scale Private 5G Open RAN Testbed," ACM SIGCOMM Computer Communication Review, Volume 55, Issue 3, pp. 19–28, 2025.
- C. Sun, **U. Pawar**, M. Khoja, X. Foukas, M. K. Marina, and B. Radunovic, "SpotLight: Accurate, Explainable and Efficient Anomaly Detection for Open RAN," ACM MobiCom, 2024.
- M. Sharma, **U. Pawar**, A. Antony Franklin, and B. R. Tamma, "Proactive Clustering of Base Stations in 5G C-RAN using Cellular Traffic Prediction," IEEE 8th International Conference on Network Softwarization (NetSoft), pp. 339–347, 2022.
- **U. Pawar**, B. R. Tamma, and A. Antony Franklin, "Traffic-Aware Compute Resource Tuning for Energy Efficient Cloud RANs," IEEE Global Communications Conference (GLOBECOM), pp. 1–6, 2021.

- **U. Pawar**, A. K. Singh, K. Malde, B. R. Tamma, and A. Antony Franklin, "Understanding Energy Consumption of Cloud Radio Access Networks: An Experimental Study," IEEE 3rd 5G World Forum (5GWF), 2020.

Workshop and Demo Papers

- **U. Pawar**, J. Yang, A. E. Ferguson, D. Mahendiran, and M. K. Marina, "Panoramic: A Cross-Layer Measurement Platform for Open RAN Deployments," 3rd ACM Workshop on Open and AI RAN (Open AI-RAN '26), 2026.
- **U. Pawar**, A. E. Ferguson, Y. Xia, X. Foukas, M. K. Marina, and B. Radunovic, "Chronos: A Cloud-Based RAN Emulation System," ACM MobiCom '26 Demos, 2026.
- **U. Pawar**, A. E. Ferguson, T. Wang, and M. K. Marina, "A Campus Scale Private 5G Open RAN Testbed," ACM MobiCom '25 Demos, pp. 1248–1250, 2025. **Best Demo Award.**
- **U. Pawar**, A. E. Ferguson, Y. Takano, J. Larrea, X. Foukas, M. K. Marina, and B. Radunovic, "Towards Scalable and Cost-Effective RAN Emulation Leveraging the Public Cloud," ACM HotMobile, pp. 43–48, 2025.
- C. Sun, **U. Pawar**, M. Khoja, X. Foukas, M. K. Marina, and B. Radunovic, "SpotLight – An Open RAN Anomaly Detection and Identification System," ACM MobiCom '24 Demos, pp. 1769–1771, 2024.

* Equal contribution.

AWARDS AND SCHOLARSHIPS

- Best Demo Award, ACM MobiCom '25 (2025)
- Travel Grant, ACM MobiCom '26 (2026)
- Ph.D. Studentship, School of Informatics, University of Edinburgh (2022 – 2026)
- Research Assistant Studentship, Department of Computer Science and Engineering, IIT Hyderabad (2019 – 2021)

TEACHING

- Marker, Computer Networks, University of Edinburgh (2022, 2024, 2026)
- Teaching Assistant, IIT Hyderabad (2020 – 2021): Networked Wireless Systems (CS5070), Internet of Things (CS5453), Network Security (CS6903), Software-Defined Networking (CS5643)